Requirements Gathering

3.1 Stakeholder Analysis

- **Student (Learner):** Requires an intuitive, friction-free portal to discover relevant educational material, execute secure payments, stream video content without buffering, track learning progress, test comprehension via quizzes, and obtain authenticated credentials.
- **Instructor (Content Creator & Educator):** Demands a rich authoring environment to structure multi-tier curricula, upload high-definition video assets, set dynamic pricing structures, interact with students via lecture Q&A, and review comprehensive enrollment and earnings metrics.
- **Platform Administrator:** Responsible for system governance, content moderation (course approval/rejection workflows), user management, category taxonomy configuration, platform-wide financial reporting, and security auditing.
- **DEPI Evaluation Board & Industry Mentors:** Assess technical execution against real-world software engineering benchmarks, clean code standards, architectural patterns, and cloud delivery.

3.2 User Stories & Acceptance Criteria

User Story 1: Course Discovery & Multi-Facet Filtering

- **As a** prospective Student,
- **I want to** search the course catalog using keywords and filter by category, skill level, rating, and price,
- **So that** I can rapidly locate courses tailored to my professional learning goals.
- **Acceptance Criteria:**
    - Search queries return relevant courses with match highlighting within 1 second.
    - Multi-facet filters update course listings dynamically without breaking pagination state.
    - Course detail cards display title, instructor name, average rating, duration, and pricing.

User Story 2: Hierarchical Course Creation & Video Ingestion

- **As an** Instructor,
- **I want to** build courses organized into sections and lessons and upload video files directly to cloud storage,
- **So that** I can publish a structured curriculum with dependable media streaming.
- **Acceptance Criteria:**
    - Instructors can add, reorder, update, and remove sections and lessons through an interactive builder.
    - Video uploads provide progress indicators and store assets in Azure Blob Storage.
    - System requires mandatory course metadata (title, description, category, price, thumbnail) prior to review submission.

User Story 3: Verified Assessment & Dynamic Certification

- **As an** enrolled Student,
- **I want to** complete section quizzes and receive a tamper-proof PDF certificate upon full course completion,
- **So that** I can demonstrate verified competence on professional networks like LinkedIn.
- **Acceptance Criteria:**
    - Quizzes evaluate answers automatically and enforce configured passing grade thresholds.
    - Reaching 100% course progress unlocks automated PDF certificate generation containing a unique verification code and public validation link.

User Story 4: Platform Quality Moderation & Approvals

- **As a** Platform Administrator,
- **I want to** inspect submitted course drafts, preview curriculum assets, and approve or reject publications with feedback,
- **So that** EduHub maintains high educational and technical quality standards.
- **Acceptance Criteria:**

- Administrator dashboard flags all courses submitted with status PendingApproval.
- Administrators can review all video lessons and trigger status updates to Published or Rejected with explanatory notes.

3.3 Functional Requirements (Expanded & Advanced)

1. User Identity, Security & Role-Based Access Control

- **FR-1.1:** Secure account registration and authentication for Students, Instructors, and Administrators using ASP.NET Core Identity.
- **FR-1.2:** Role-based and claim-based access controls to protect administrative and instructor resources.
- **FR-1.3:** Self-service password recovery, email confirmation workflows, and secure session management.
- **FR-1.4:** User profile management enabling biography updates, avatar image uploads, social links, and security credential changes.

2. Course Catalog & Curriculum Management

- **FR-2.1:** Full CRUD operations for courses, structured hierarchically into Sections (modules) and Lessons (units).
- **FR-2.2:** Support for multi-level taxonomic categorization (Categories and Subcategories) with metadata tagging.
- **FR-2.3:** Course state machine governing lifecycle transitions: Draft, PendingApproval, Published, and Archived.
- **FR-2.4:** Rich text description editing with embedded learning outcomes, target audience, and prerequisite definitions.
- **FR-2.5:** Course resource attachments supporting supplemental materials (PDF guides, code repositories, reference sheets).

3. Media Ingestion & Secure Video Delivery

- **FR-3.1:** Direct-to-cloud video uploading integrated with Microsoft Azure Blob Storage.
- **FR-3.2:** Generation of short-lived, encrypted Shared Access Signature (SAS) URLs to restrict direct media access to authorized, enrolled learners.

- **FR-3.3:** Responsive custom video player powered by Video.js featuring speed control, theater mode, autoplay, and keyboard navigation.
- **FR-3.4:** Automatic duration extraction and resolution metadata capture upon media processing.

4. Interactive Learning Engine & Progress Telemetry

- **FR-4.1:** Automated lesson progress tracking marking units completed upon full video playback or manual acknowledgment.
- **FR-4.2:** Dynamic visual progress bars showing section-level and overall course completion percentages.
- **FR-4.3:** In-player timestamped personal note-taking allowing students to record notes tied to specific video moments.
- **FR-4.4:** Bookmark system enabling learners to flag critical lectures for quick reference.
- **FR-4.5:** Lecture-specific Q&A discussion board enabling threaded communication between students and instructors.

5. Assessment, Examination & Student Evaluation

- **FR-5.1:** Instructor quiz builder supporting multiple question formats: Multiple Choice (Single Answer), Multiple Choice (Multiple Answers), and True/False.
- **FR-5.2:** Configurable assessment rules including customizable passing thresholds (e.g., 80%), attempt limits, and randomized question distribution.
- **FR-5.3:** Instant automated grading engine with question-by-question answer explanations displayed upon submission.
- **FR-5.4:** Comprehensive quiz attempt audit history tracking historical scores and completion timestamps.

6. Social Proof, Reviews & Feedback System

- **FR-6.1:** Course rating system allowing verified enrolled learners to submit 1 to 5 star ratings with written reviews.
- **FR-6.2:** Weighted calculation of aggregate course ratings displayed alongside rating breakdown metrics.
- **FR-6.3:** Instructor response capabilities to student feedback.

- **FR-6.4:** Administrative moderation tools to flag, hide, or address malicious reviews.

7. E-Commerce, Multi-Item Checkout & Payments

- **FR-7.1:** Full shopping cart supporting single-course enrollment or multi-course batch purchases.
- **FR-7.2:** End-to-end integration with Stripe Checkout supporting credit/debit card transactions and real-time webhook handling.
- **FR-7.3:** Promotional coupon engine utilizing the **Strategy Pattern** to execute Percentage-off, Fixed-amount-off, and Free enrollment campaigns.
- **FR-7.4:** Automatic generation of itemized purchase invoices and order history tracking accessible via student profiles.
- **FR-7.5:** Automated instructor revenue-share calculations logging net platform fee deductions and instructor balances.

8. Automated Certification & Credential Verification

- **FR-8.1:** Dynamic server-side compilation of PDF completion certificates using iTextSharp upon 100% course requirement fulfillment.
- **FR-8.2:** Injection of personalized metadata: Student Full Name, Course Title, Completion Date, Instructor Signature, and Unique Certificate Hash.
- **FR-8.3:** Embedded QR code linking directly to a public verification endpoint.
- **FR-8.4:** Public verification portal enabling employers to confirm certificate validity using the certificate GUID without authentication.

9. AI-Assisted Pedagogical Tools

- **FR-9.1:** Automated lecture summarization generating key bullet points and takeaways from lesson content.
- **FR-9.2:** AI-assisted quiz generator helping instructors generate assessment questions based on course learning objectives.
- **FR-9.3:** Semantic course search indexing course syllabi to locate specific lessons based on conceptual queries.

10. Administrative Governance, Analytics & Reporting

- **FR-10.1:** Centralized administrative control center with granular oversight of users, courses, financial transactions, and content flags.

- **FR-10.2:** Course approval workflow allowing administrators to review, approve, or reject new courses with line-item review comments.
- **FR-10.3:** Interactive analytics and visual metrics powered by Chart.js displaying enrollment trends, platform revenue, top-selling courses, and student drop-off rates.
- **FR-10.4:** Data export capabilities allowing instructors and administrators to export financial ledgers and enrollment rosters to CSV and Excel.

3.4 Non-Functional Requirements (Enterprise-Grade)

1. Performance & Throughput

- **NFR-1.1 (Latency):** Standard web pages must render with a Time to First Byte (TTFB) below 800ms; catalog API endpoints must respond within 500ms under standard loads.
- **NFR-1.2 (Concurrency):** System must sustain up to 500 concurrent active users browsing, taking quizzes, and updating progress without exceeding 70% CPU/Memory utilization.
- **NFR-1.3 (Database Optimization):** Database access must utilize indexed lookups, explicit pagination, and eager loading (Include()) where appropriate to prevent N+1 query patterns.
- **NFR-1.4 (Caching Strategy):** Static resources, platform settings, and course categories must leverage in-memory caching (IMemoryCache) to minimize redundant database round-trips.

2. Information Security & Compliance

- **NFR-2.1 (OWASP Top 10 Mitigation):** Strict defenses against common web vulnerabilities:
    - *SQL Injection:* Completely mitigated through parameterized queries via Entity Framework Core.
    - *Cross-Site Scripting (XSS):* Automatic HTML encoding enforced by Razor View Engine, combined with input sanitization.
    - *Cross-Site Request Forgery (CSRF):* Mandatory validation of [ValidateAntiForgeryToken] across all non-GET actions.

- **NFR-2.2 (Data Protection in Transit and at Rest):** Full enforcement of HTTPS/TLS 1.3 across all communication channels; database fields holding sensitive keys must be encrypted at rest.
- **NFR-2.3 (Credential Safety):** User passwords must be salted and hashed using PBKDF2 with SHA-256 via ASP.NET Identity.
- **NFR-2.4 (Media Protection):** Direct access to raw storage blobs must be blocked; media is only accessible through short-lived SAS tokens valid for single-session windows.

3. Reliability, Availability & Fault Tolerance

- **NFR-3.1 (High Availability):** Platform uptime targeted at a minimum of 99.9% utilizing Azure App Service deployment slots and automated health checks.
- **NFR-3.2 (Data Integrity & ACID Guarantees):** All financial settlements, enrollments, and status updates must execute within explicit Unit of Work database transactions to eliminate partial data writes.
- **NFR-3.3 (Automated Recovery):** Global exception handling middleware must catch unhandled runtime errors, display friendly error views to users, and log full stack traces securely to Azure Application Insights.

4. Scalability & Architectural Maintainability

- **NFR-4.1 (Decoupled Layering):** Strict architectural isolation across Presentation (EduHub.Web), Business Logic (EduHub.Services), and Data Persistence (EduHub.Data), preventing circular dependencies.
- **NFR-4.2 (Stateless Application Design):** Web application layer must maintain a stateless architecture (session storage backed by distributed memory or clean cookie tokens) to allow zero-downtime horizontal scaling across multiple cloud instances.
- **NFR-4.3 (Maintainable Code Base):** Source code must follow standard C# coding conventions, SOLID principles, and meaningful naming patterns with public interfaces for all business services.

5. Usability, Responsiveness & Accessibility

- **NFR-5.1 (Responsive Design):** Web layouts must be fully responsive across mobile (>= 320px), tablet (>= 768px), and desktop (>= 1200px) displays using Bootstrap 5 grid systems.

- **NFR-5.2 (Accessibility):** User interfaces must comply with WCAG 2.1 Level AA accessibility standards, maintaining appropriate contrast ratios, keyboard navigation for the media player, and descriptive aria-labels.
- **NFR-5.3 (Intuitive UX):** Form submissions must feature clear client-side validation (jQuery Validation) and immediate toast notifications (Toastr JS) for background operations.